

Hadron/QCD cluster — the clean results are RATIOS, the absolutes are limited (Grace, 2026-06-29). Opening the hadron lane (paired with Elie) with the Cal #286 discipline lens.

CLEAN target-innocent results (all dimensionless substrate ratios): (1) meson STRANGE-step $f(+s)/f = C_2/n_C = 6/5$, universal across $\pi \rightarrow K$, $D \rightarrow D_s$, $B \rightarrow B_s$ (3 spectator-independent confirmations, $\sim 1\%$, INV-5677); (2) neutron magnetic moment $\mu_n = -C_2/\pi = -6/\pi$ (0.17%, INV-5678), $\mu_p/\mu_n = -N_c/\text{rank} = -3/2$ (constituent QM); (3) $f_K/f_\pi = C_2/n_C$ (corpus, = the strange-step). FISH-SUSPECT (per-number fits, flagged NOT cited as derivations): the recurring ‘13’ (p - n split $91/36 = g_{13}/C_2^2$; $m_\pi = N_{cg} g_{13} m_e$; $g_{\pi NN} = 1325/24$; $13 = C_2 + g$ but used as a fit) and ‘47’ ($m_p/m_\pi = 47/g$).

SCHEME/CONVENTION-LIMITED (honest negatives): the p - n splitting ABSOLUTE (EM + scheme-dependent quark-mass dm),

m_π ABSOLUTE ($273m_e$ = substrate-product fit, multiple factorizations $\sqrt{3050} = 37 \cdot 13$), the heavy-quark (c,b) decay-constant steps. META-PATTERN: the CLEAN hadron observables are dimensionless RATIOS / substrate-ratio structure (C_2/n_C , C_2/π , N_c/rank); the ABSOLUTE masses/numbers are scheme/convention/fish-limited – the SAME structure-clean/scale-limited split as the gauge ($b_3 = g$ clean, couplings running) and fermion (ratios clean, absolutes scheme-limited) sectors. This is the universal BST pattern and the ‘why’ of the 9/26 count saturation: the substrate fixes dimensionless STRUCTURE cleanly, not absolute SCALE. C_2/n_C is a Cal #35 cross-role (meson SU(3)-breaking + nuclear κ ’s). No count move (identified/structural).

Grace

2026-06-29 Monday

Hadron/QCD cluster: ratios clean, scales limited

Hadron lane (paired with Elie), Cal #286 lens armed. The corpus hadron sector has the same numerology tell as the magic numbers — clean substrate ratios hiding under per-number fits.

CLEAN (target-innocent, all dimensionless ratios)

result	form	precision
meson strange-step	$f(+s)/f = C_2/n_C$ ($\pi \rightarrow K, D \rightarrow Ds, B \rightarrow Bs$ — 3 spectators)	~1%, robust (INV-5677)
neutron moment	$\mu_n = -C_2/\pi$	0.17% (INV-5678)
moment ratio	$\mu_p/\mu_n = -N_c/\text{rank} = -3/2$ (constituent QM)	2.7%
f_K/f_π	C_2/n_C (corpus = the strange-step)	0.17%

FISH-SUSPECT (per-number fits — flagged, NOT derivations)

- “13”: p-n split $91/36 = g \cdot 13/C_2^2$; $m_\pi = N_c \cdot g \cdot 13 \cdot m_e$; $g_{\pi NN} = 13 \cdot 25/24$. (13 = $C_2 + g$, but used as a fit.)
- “47”: $m_p/m_\pi = 47/g$. Not a clean primary. These are the hadron analog of the magic-number per-number forms (Elie’s catch).

SCHEME/CONVENTION-LIMITED (honest negatives = wins)

- p-n splitting absolute — EM self-energy + scheme-dependent ($m_d - m_u$). The mechanism (EM + dm) is real; the absolute inherits the quark-mass scheme-dependence. NOT the 91/36 fit.
- m_π absolute — $273 \cdot m_e$ is a substrate-product fit with multiple factorizations ($\sqrt{30 \cdot 50} = 3 \cdot 7 \cdot 13$); the chiral handle is $\sqrt{(n_C(n_C+1))}$, but the absolute is not clean.
- heavy-quark (c,b) decay-constant steps — convention-dependent.

The meta-pattern (the “why” of count saturation)

Every CLEAN hadron result is a dimensionless RATIO / substrate-ratio structure ($C_2/n_C, C_2/\pi, N_c/\text{rank}$); every ABSOLUTE mass/number is scheme/convention/fish-limited. This is the SAME split as the gauge sector ($b_3 = g$ clean, couplings running) and the fermion sector (ratios clean, absolutes scheme-limited). It is the universal BST pattern: the substrate fixes dimensionless STRUCTURE cleanly, not absolute SCALE — which is exactly why the honest count saturates at 9/26 (the forced dimensionless params banked; the absolute scales running/scheme/convention-limited, the no-fishing line). C_2/n_C is a Cal #35 cross-role (meson SU(3)-breaking + nuclear κ ’s — one ratio, two sectors).

— Grace, 2026-06-29 Monday. Hadron lane: clean = ratios (strange-step C_2/n_C 3-meson, $\mu_n = -C_2/\pi$); fish = 13/47; scale-limited = p-n/ m_π absolutes. Meta: ratios clean, scales limited (universal pattern, the count-saturation reason). No count move.